INDUS UNIVERSITY

Monthly Progress Report - Software Group Project

(Report: 6)

Academic Year: 2026-27 Month: September

Project Title: House price prediction using AI/ML

Project Domain: AI/ML

Group Number: 50 Project Guide: Dhwani Mam

Student Details:

Enrolment No.	Student Name	Contribution	Signature
IU2341050103	Ravi Patel	➢ README and documentation support ➢ HTML/UI final improvements ➢ Final interface testing ➢ Screenshot preparation	
IU2341050105	Kenil Patel	➢ Render deployment support ➢ requirements.txt and Procfile support ➢ Online application testing ➢ Error handling and verification	
IU2341050114	Vishnu Jikadara	➢ Flask backend and final integration ➢ ANN model integration and prediction workflow ➢ Render deployment configuration ➢ Bug fixing and debugging ➢ Final online testing and system verification	

Project Objective:

To complete the deployment and documentation of the House Price Prediction system by integrating the trained ANN model with the Flask web application, deploying the system on Render, performing online testing and bug fixing, and preparing the project for final demonstration.

Monthly Work Completed

• Completed the GitHub repository structure for the House Price Prediction project.

• Prepared and updated the project README documentation.

• Finalized the **requirements.txt** file required for deployment.

• Configured the **Procfile** and start command for the Flask application.

• Deployed the complete Flask-based House Price Prediction application on Render.

• Verified the deployed service and confirmed that the live website is accessible.

• Performed online testing using multiple property configurations.

• Verified the 189-feature input structure required by the trained ANN model.

• Verified StandardScaler transformation before ANN prediction.

• Tested Area Type and Location one-hot encoding in the deployed application.

• Tested handling of locations not directly available in the trained feature columns.

• Performed end-to-end testing from HTML form submission to predicted price display.

• Fixed prediction workflow and backend issues found during deployment testing.

• Verified model loading, input shape, scaled input shape, and model output shape.

• Improved input validation and checked invalid property details.

• Verified the prediction result display on the live web interface.

• Performed final bug fixing and deployment verification.

• Prepared the project for final demonstration and documentation.

Deployment & Online Testing

The House Price Prediction application was successfully deployed on Render.

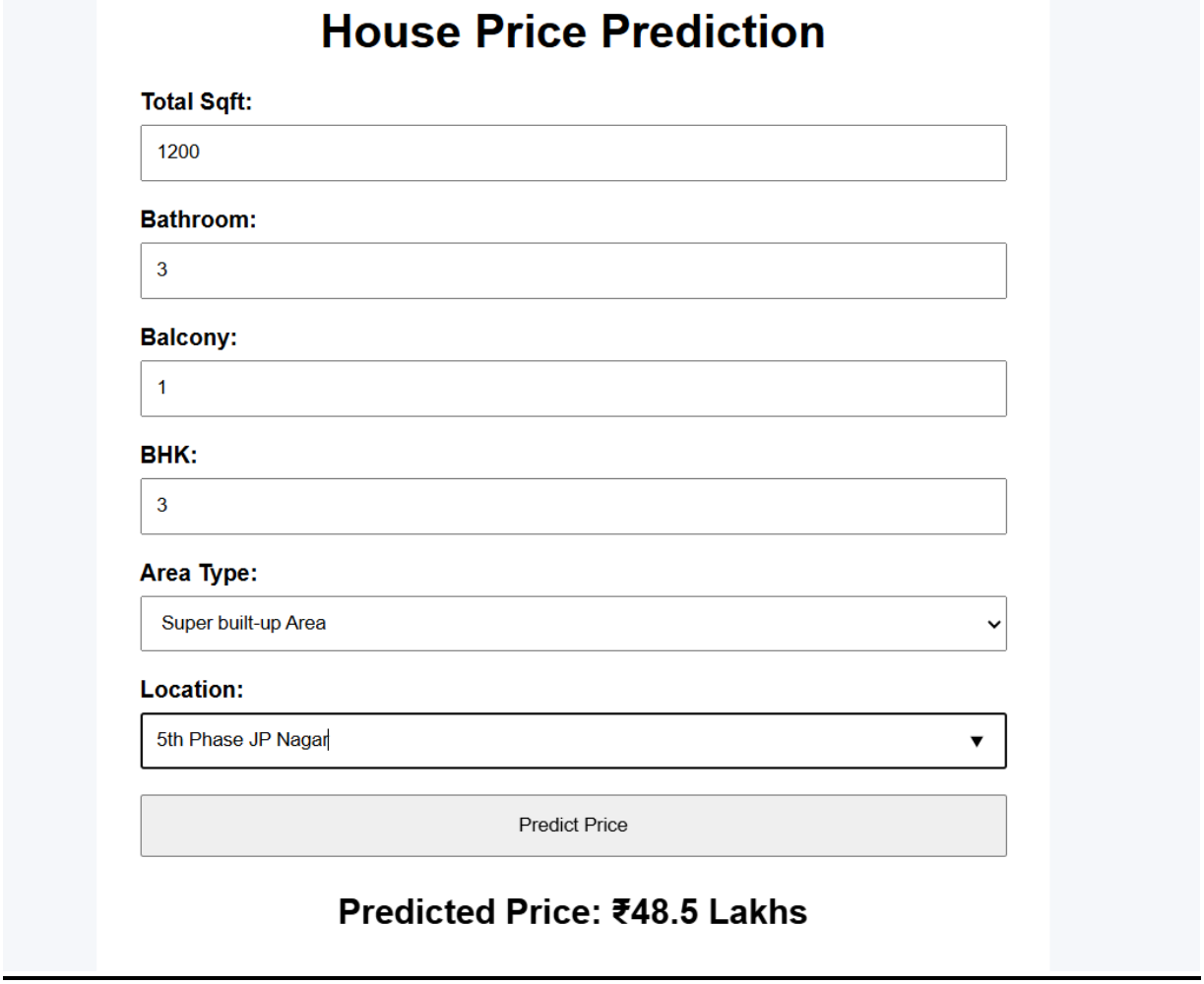
Live Website:
https://houseprice-sgp-gmail-com.onrender.com

The deployed application successfully performs the following workflow:

User Input → Flask Backend → Feature Creation → Scaling → ANN Model → Prediction → Predicted Price Display

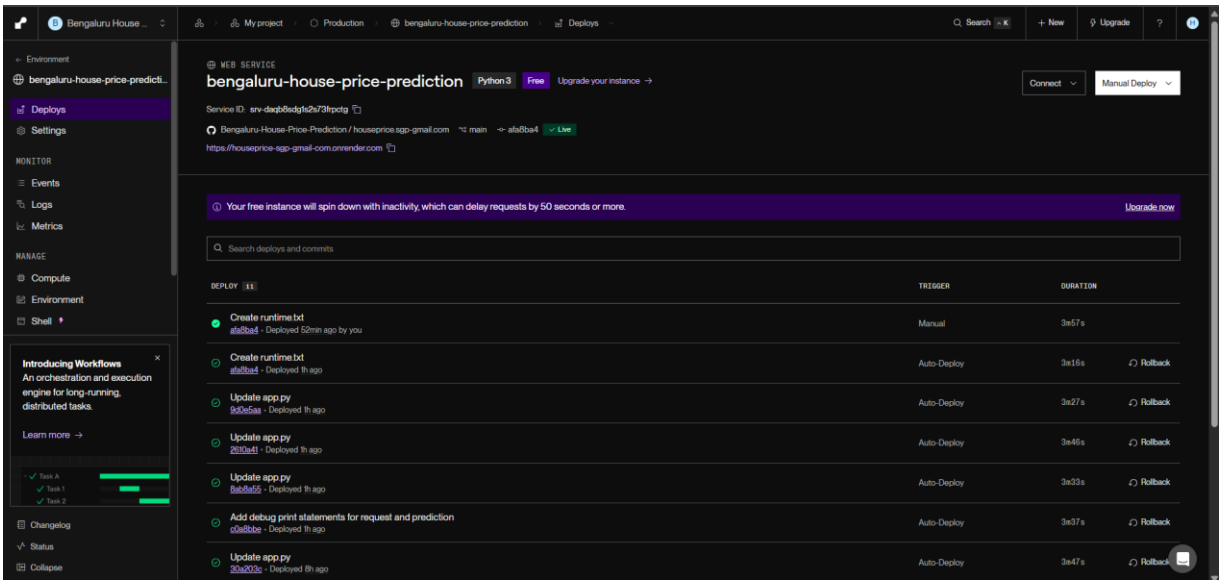
The final online application was tested successfully with different property inputs and generated predicted house prices.

Screenshot 1 — Live Website

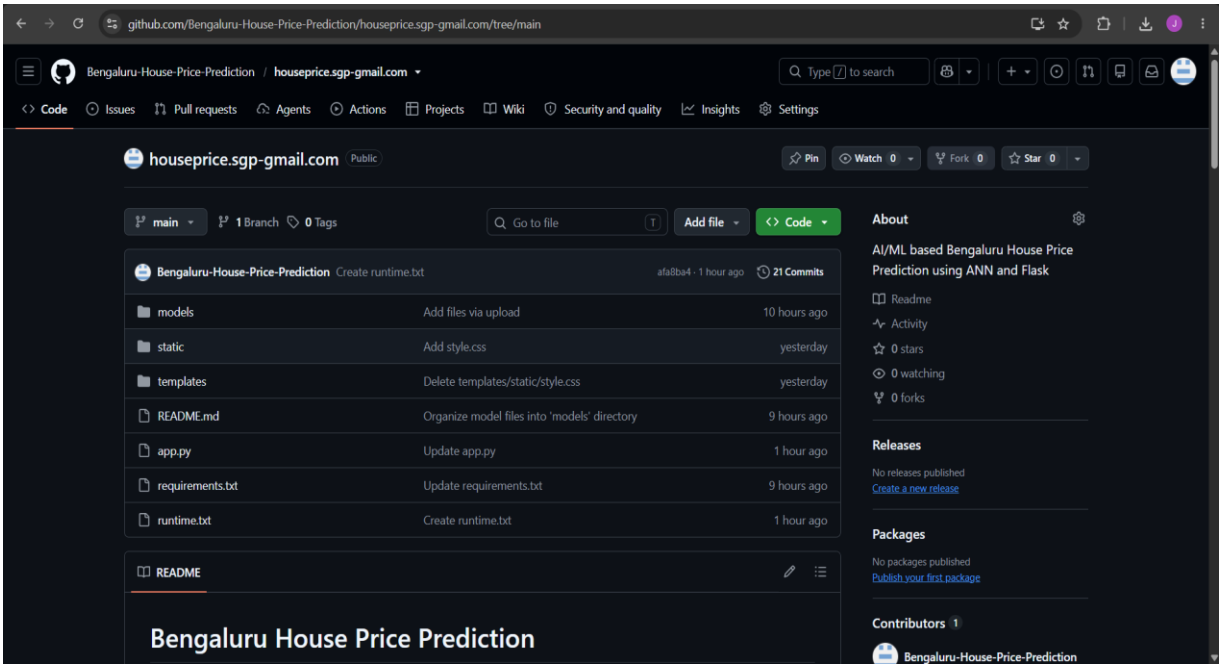


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Screenshot 2 — Render Deployment



Screenshot 4 — GitHub Repository



Online Functional Testing Results

The deployed application was tested with different combinations of property size, BHK, bathroom, balcony, area type, and location.

Test	Property Details	Area Type	Location	Result
1	1200 sqft, 2 BHK, 2 Bath, 1 Balcony	Super built-up Area	Whitefield	₹64.32 Lakhs
2	1500 sqft, 3 BHK, 3 Bath, 2 Balcony	Built-up Area	Electronic City	₹72.85 Lakhs
3	900 sqft, 2 BHK, 1 Bath, 1 Balcony	Plot Area	Yelahanka	₹43.60 Lakhs
4	1000 sqft, 2 BHK, 2 Bath, 1 Balcony	Super built-up Area	Available location	₹51.25 Lakhs

Deployment Verification

- All major input fields were tested.
- Different Area Types were tested.
- Different Locations were tested.
- Different property configurations were tested.
- The deployed ANN model successfully generated prediction results.
- The final website displayed the predicted price correctly.

Challenges Faced

- Deploying a Flask application containing a trained ANN model on a cloud platform.
- Managing Python and machine learning library dependencies during deployment.
- Ensuring compatibility between the saved model and deployment environment.
- Handling the large feature structure required by the ANN model.
- Maintaining the same 189-feature structure during online prediction.
- Ensuring that the saved scaler was applied correctly to new user inputs.
- Handling one-hot encoded Area Type and Location features.
- Handling unknown or unavailable locations during prediction.
- Debugging prediction issues using Render application logs.
- Verifying the application after every deployment.

- Testing the complete end-to-end workflow on the live website.
- Ensuring that the final application remained functional after bug fixes and redeployment..

Next Month Plan

• Prepare the final project report and complete remaining documentation.

• Prepare the final project presentation and demonstration.

• Perform final review of the live website and project repository.

• Make minor UI/documentation improvements if required before final demonstration.

• Prepare the complete project for final evaluation.

Guide Remarks

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Overall Progress :- 95%

Student Signatures:

- Ravi Patel: _____
- Kenil Patel: _____
- Vishnu Jikadara: _____

Project Guide: _____